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United States Patent	12414408
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chen; Yi-Hong et al.

Light emitting device having third electrode

Abstract

A light emitting device includes a semiconductor structure, an insulating layer, a first electrode, a second electrode, and a third electrode. The semiconductor structure includes a first type semiconductor layer, a second type semiconductor layer, and an active layer disposed between the first type semiconductor layer and the second type semiconductor layer. The insulating layer is disposed on the semiconductor structure. The first electrode is electrically connected to the first type semiconductor layer. The second electrode is electrically connected to the second type semiconductor layer. The first electrode, the second electrode, and the third electrode are structurally separated. The third electrode at least has a first portion. The first portion of the third electrode is disposed on a side wall of the semiconductor structure, and the insulating layer is located between the third electrode and the semiconductor structure.

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Appl. No.: 17/890285

Filed: August 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230111295 A1	Apr. 13, 2023

Foreign Application Priority Data

TW	110137822	Oct. 12, 2021
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Publication Classification

Int. Cl.: H10H20/831 (20250101); H10H20/80 (20250101); H10H20/84 (20250101); H10H20/855 (20250101)

U.S. Cl.:

CPC H10H20/8314 (20250101); H10H20/84 (20250101); H10H20/855 (20250101); H10H20/882 (20250101)

Field of Classification Search

CPC: H01L (33/385); H01L (33/44); H01L (33/58)

USPC: 257/98

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 110137822, filed on Oct. 12, 2021. The entirety of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a light emitting device.

Description of Related Art

(3) A light emitting diode display apparatus includes a driving backplane and multiple light emitting diode chips transposed onto the driving backplane. Inheriting the characteristics of light emitting diodes, light emitting diode display apparatuses have the advantages of power saving, high efficiency, high brightness, and fast response time. Compared with organic light emitting diode display apparatuses, light emitting diode display apparatuses also have the advantages of easy color adjustment, long light emitting lifetime, and no image burn-in. Therefore, the light emitting diode display apparatus is regarded as the next generation display technology. However, a light emitting diode chip is etched from an epitaxial stack; during the etching process, the light emitting diode chip is damaged and produce defects, which leads to a high ratio of side wall current leakage and non-radiative recombination of the light emitting diode chip, which affects the luminous efficiency of the light emitting diode chip.

SUMMARY

(4) The disclosure provides a light emitting device with high luminous efficiency.

(5) The light emitting device of the disclosure includes a semiconductor structure, an insulating layer, a first electrode, a second electrode, and a third electrode. The semiconductor structure includes a first type semiconductor layer, a second type semiconductor layer, and an active layer disposed between the first type semiconductor layer and the second type semiconductor layer. The insulating layer is disposed on the semiconductor structure. The first electrode is electrically connected to the first type semiconductor layer. The second electrode is electrically connected to the second type semiconductor layer. The first electrode, the second electrode, and the third electrode are structurally separated. The third electrode at least has a first portion. The first portion of the third electrode is disposed on a side wall of the semiconductor structure, and the insulating layer is located between the third electrode and the semiconductor structure.

(6) In an embodiment of the disclosure, the side wall of the semiconductor structure includes a side wall of the active layer, and the first portion of the third electrode is disposed on the side wall of the active layer.

(7) In an embodiment of the disclosure, the side wall of the semiconductor structure further includes a side wall of the first type semiconductor layer, and the first portion of the third electrode is further disposed on the side wall of the first type semiconductor layer.

(8) In an embodiment of the disclosure, the side wall of the semiconductor structure further includes a side wall of the second type semiconductor layer, and the first portion of the third electrode is further disposed on the side wall of the second type semiconductor layer.

(9) In an embodiment of the disclosure, the light emitting device further includes a first conductive pattern, a second conductive pattern, and a third conductive pattern. The first conductive pattern is disposed on the first electrode and is electrically connected to the first electrode. The second conductive pattern is disposed on the second electrode and is electrically connected to the second electrode. The third conductive pattern is disposed on the third electrode and is electrically connected to the first portion of the third electrode. The first conductive pattern, the second

conductive pattern, and the third conductive pattern are structurally separated.

(10) In an embodiment of the disclosure, the semiconductor structure has a first surface and a second surface opposite to each other. The side wall of the semiconductor structure is disposed between the first surface and the second surface. The third electrode further has a second portion disposed on the first surface of the semiconductor structure and electrically connected to the first portion of the third electrode. The third conductive pattern is disposed on the second portion of the third electrode.

(11) In an embodiment of the disclosure, the light emitting device further includes at least one first optical microstructure overlapping the active layer.

(12) In an embodiment of the disclosure, the active layer emits a light beam, and the at least one first optical microstructure converges the light beam.

(13) In an embodiment of the disclosure, the active layer has a first region and a second region. The second region of the active layer is located between the first region of the active layer and the side wall of the semiconductor structure. The light emitting device further includes at least one second optical microstructure. The at least one first optical microstructure and the at least one second optical microstructure respectively overlap the first region and the second region of the active layer.

(14) In an embodiment of the disclosure, the active layer emits a light beam, the at least one first optical microstructure converges the light beam, and the at least one second optical microstructure diverges the light beam.

(15) In an embodiment of the disclosure, the light emitting device further includes at least one second optical microstructure overlapping the active layer. The active layer emits a light beam, and the at least one second optical microstructure diverges the light beam.

(16) In an embodiment of the disclosure, the third electrode has a reverse bias.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) FIG. 1 is a cross-sectional schematic view of a light emitting device **10** according to an embodiment of the disclosure.

(2) FIG. 2 is a bottom schematic view of the light emitting device **10** of FIG. 1.

(3) FIG. 3 shows the internal energy band of the light emitting device **10** according to an embodiment of the disclosure when a third electrode **160** is not biased.

(4) FIG. 4 shows the internal energy band of the light emitting device **10** according to an embodiment of the disclosure when the third electrode **160** has a reverse bias.

(5) FIG. 5 is a cross-sectional schematic view of a light emitting device **10A** according to an embodiment of the disclosure.

(6) FIG. 6 is a cross-sectional schematic view of a light emitting device **10B** according to an embodiment of the disclosure.

(7) FIG. 7 is a cross-sectional schematic view of a light emitting device **10C** according to an embodiment of the disclosure.

(8) FIG. 8 is a cross-sectional schematic view of a light emitting device **10C** according to an embodiment of the disclosure.

(9) FIG. 9 is a cross-sectional schematic view of a light emitting device **10D** according to an embodiment of the disclosure.

(10) FIG. 10 is a cross-sectional schematic view of a light emitting device **10E** according to an embodiment of the disclosure.

(11) FIG. 11 is a cross-sectional schematic view of a light emitting device **10F** according to an embodiment of the disclosure.

(12) FIG. 12 is a cross-sectional schematic view of a light emitting device **10G** according to an

embodiment of the disclosure.

(13) FIG. 13 is a cross-sectional schematic view of a light emitting device 10H according to an embodiment of the disclosure.

(14) FIG. 14 is a cross-sectional schematic view of a light emitting device 10I according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

(15) References of the exemplary embodiments of the disclosure are to be made in detail. Examples of the exemplary embodiments are illustrated in the drawings. If applicable, the same reference numerals in the drawings and the descriptions indicate the same or similar parts.

(16) It should be understood that when an element such as a layer, a film, an area, or a substrate is indicated to be “on” another element or “connected to” another element, it may be directly on another element or connected to another element, or an element in the middle may exist. In contrast, when an element is indicated to be “directly on another element” or “directly connected to” another element, an element in the middle does not exist. As used herein, “to connect” may indicate to physically and/or electrically connect. Furthermore, “to electrically connect” or “to couple” may also be used when other elements exist between two elements.

(17) The usages of “approximately”, “similar to”, or “substantially” indicated throughout the specification include the indicated value and an average value having an acceptable deviation range, which is a certain value confirmed by people skilled in the art, and is a certain amount considered the discussed measurement and measurement-related deviation (that is, the limitation of measurement system). For example, “approximately” may indicate to be within one or more standard deviations of the indicated value, or being within $\pm 30\%$, $\pm 20\%$, $\pm 10\%$, $\pm 5\%$. Furthermore, the usages of “approximately”, “similar to”, or “substantially” indicated throughout the specification may refer to a more acceptable deviation scope or standard deviation depending on optical properties, etching properties, or other properties, and all properties may not be applied with one standard deviation.

(18) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as that commonly understood by one of ordinary skill in the art to which this disclosure belongs. It should be further understood that terms such as those defined in commonly used dictionaries should be construed as having meanings consistent with their meanings in the context of the related art and the disclosure, and are not to be construed as idealized or excessive formal meaning, unless expressly defined as such herein.

(19) FIG. 1 is a cross-sectional schematic view of a light emitting device 10 according to an embodiment of the disclosure. FIG. 2 is a bottom schematic view of the light emitting device 10 of FIG. 1.

(20) Referring to FIG. 1 and FIG. 2, the light emitting device 10 includes a semiconductor structure S. The semiconductor structure S includes a first type semiconductor layer 110, a second type semiconductor layer 120, and an active layer 130. The active layer 130 is disposed between the first type semiconductor layer 110 and the second type semiconductor layer 120. The semiconductor structure S has a first surface Sa, a second surface Sb, and a side wall Sc. The first surface Sa and the second surface Sb are disposed opposite to each other, and the side wall Sc is disposed between the first surface Sa and the second surface Sb.

(21) In this embodiment, an epitaxial stack (not shown) includes a first type semiconductor material layer (not shown), an active material layer (not shown), and a second type semiconductor material layer (not shown) for forming the first type semiconductor layer 110, the active layer 130, and the second type semiconductor layer 120, respectively. The semiconductor structure S is etched from the epitaxial stack, and the side wall Sc of the semiconductor structure S may refer to the etched surface of the semiconductor structure S. In this embodiment, the side wall Sc (i.e., the etched surface) of the semiconductor structure S includes a side wall 110a of the first type semiconductor layer 110, a side wall 130a of the active layer 130, and a side wall 120a of the

second type semiconductor layer **120**.

(22) For example, in this embodiment, the side wall **110a** of the first type semiconductor layer **110** may include a first sub-side wall **110a-1** and a second sub-side wall **110a-2**. The first sub-side wall **110a-1** is connected to the second sub-side wall **110a-2**, the first sub-side wall **110a-1** and the second sub-side wall **110a-2** form an included angle θ , the side wall **130a** of the active layer **130** is connected to the second sub-side wall **110a-2** of the first type semiconductor layer **110**, and the side wall **120a** of the second type semiconductor layer **120** is connected to the side wall **130a** of the active layer **130**. In other words, in this embodiment, the side wall Sc of the semiconductor structure S formed by connecting the side wall **110a** of the first type semiconductor layer **110**, the side wall **130a** of the active layer **130**, and the side wall **120a** of the second type semiconductor layer **120** may not be located on the same plane, in which the side wall Sc of the semiconductor structure S may selectively have a level difference, but the disclosure is not limited thereto.

(23) The light emitting device **10** further includes an insulating layer **140** disposed on the semiconductor structure S. Specifically, in this embodiment, the insulating layer **140** covers at least the side wall Sc and the first surface Sa of the semiconductor structure S, and has a first opening **140a** and a second opening **140b** respectively overlapping the first type semiconductor layer **110** and the second type semiconductor layer **120**. For example, in this embodiment, the material of the insulating layer **140** may be an inorganic material (e.g., silicon oxide, silicon nitride, silicon oxynitride, or a stacked layer of at least two of the foregoing materials), an organic material, or a combination thereof.

(24) The light emitting device **10** further includes a first electrode **151** and a second electrode **152** electrically connected to the first type semiconductor layer **110** and the second type semiconductor layer **120**, respectively. Specifically, in this embodiment, the first electrode **151** and the second electrode **152** are respectively filled in the first opening **140a** and the second opening **140b** of the insulating layer **140** to be electrically connected to the first type semiconductor layer **110** and the second type semiconductor layer **120**, respectively. For example, in this embodiment, the first type semiconductor layer **110**, the second type semiconductor layer **120**, and the active layer **130** may be an n-type semiconductor layer, a p-type semiconductor layer, and a multiple quantum well layer, respectively, and the first electrode **151** and the second electrode **152** may be the cathode and the anode of the light emitting device **10**, respectively, but the disclosure is not limited thereto.

(25) The light emitting device **10** further includes a third electrode **160**. The third electrode **160** is disposed on the insulating layer **140**. The insulating layer **140** is located between the third electrode **160** and the semiconductor structure S. The first electrode **151**, the second electrode **152**, and the third electrode **160** are structurally separated. The third electrode **160** at least has a first portion **161** disposed on the side wall Sc of the semiconductor structure S.

(26) In this embodiment, the first portion **161** of the third electrode **160** is disposed at least on the side wall Sc of the semiconductor structure S near the first surface Sa. In this embodiment, the first portion **161** of the third electrode **160** is disposed at least on the side wall **130a** of the active layer **130**. For example, in this embodiment, the first portion **161** of the third electrode **160** may be selectively disposed on the side wall **110a** of the first type semiconductor layer **110** and on the side wall **120a** of the second type semiconductor layer **120** in addition to being disposed on the side wall **130a** of the active layer **130**, but the disclosure is not limited thereto. In addition, in this embodiment, the third electrode **160** may optionally have a second portion **162** disposed on the first surface Sa of the semiconductor structure S and electrically connected to the first portion **161** of the third electrode **160**, but the disclosure is not limited thereto.

(27) FIG. 3 shows the internal energy band of the light emitting device **10** according to an embodiment of the disclosure when a third electrode **160** is not biased. FIG. 4 shows the internal energy band of the light emitting device **10** according to an embodiment of the disclosure when the third electrode **160** has a reverse bias.

(28) Referring to FIG. 1 and FIG. 3, when the third electrode **160** is not biased, the side wall Sc

(i.e., the etched surface) of the semiconductor structure S has defects due to the etching process, so that the energy band of the light emitting device **10** is deformed near the interface between the side wall Sc of the semiconductor structure S and the insulating layer **140**, and electrons e⁻ are easily trapped near the side wall Sc, thereby non-radiative recombination with electron holes occurs near the side wall Sc, resulting in a leakage current i and a loss in luminous efficiency in the light emitting device **10**.

(29) Referring to FIG. 1 and FIG. 4, it is worth mentioning that when the third electrode **160** has a reverse bias, the deformation of the energy band of the light emitting device **10** near the interface between the side wall Sc and the insulating layer **140** may be leveled, so that the electrons e⁻ are not easily trapped near the side wall Sc, thereby radiatively recombining with the electron holes inside the active layer **130**. Thereby, the luminous efficiency of the light emitting device **10** may be improved.

(30) Referring to FIG. 1 and FIG. 2, in this embodiment, the light emitting device **10** further includes a first conductive pattern **171**, a second conductive pattern **172**, and a third conductive pattern **173**. The first conductive pattern **171** is disposed on the first electrode **151** and is electrically connected to the first electrode **151**. The second conductive pattern **172** is disposed on the second electrode **152** and is electrically connected to the second electrode **152**. The third conductive pattern **173** is disposed on the third electrode **160** and is electrically connected to the first portion **161** of the third electrode **160**. The first conductive pattern **171**, the second conductive pattern **172**, and the third conductive pattern **173** are structurally separated.

(31) When the light emitting device **10** is transposed onto the driving backplane (not shown), the first conductive pattern **171**, the second conductive pattern **172**, and the third conductive pattern **173** connect the first electrode **151**, the second electrode **152**, and the third electrode **160** of the light emitting device **10** and the driving backplane, thereby forming a display apparatus.

(32) For example, in this embodiment, the third electrode **160** may selectively include the second portion **162** located on the first surface Sa of the semiconductor structure S, and the third conductive pattern **173** may be selectively disposed on the second portion **162** of the third electrode **160** to be electrically connected with the first portion **161** of the third electrode **160**. However, the disclosure is not limited thereto, and in other embodiments, the third conductive pattern **173** may also be disposed at other positions.

(33) Referring to FIG. 1, in this embodiment, the first electrode **151** and the second electrode **152** of the light emitting device **10** are disposed on the same side of the active layer **130**; after the light emitting device **10** is transposed and electrically connected to the driving backplane (not shown), the first conductive pattern **171** and the second conductive pattern **172** electrically connect the light emitting device **10** and the driving backplane are disposed between the first electrode **151** and the second electrode **152** of the light emitting device **10** and the driving backplane, and the light emitting device **10** may be a flip chip light emitting diode (flip chip LED). However, the disclosure is not limited thereto, and in other embodiments, the light emitting device **10** may also be other forms of light emitting diode devices, which are illustrated in the following paragraphs with other drawings.

(34) It is to be noted that the following embodiments use the reference numerals and a part of the contents of the above embodiments, and the same reference numerals are used to denote the same or similar elements, and the description of the same technical contents is omitted. For the description of the omitted part, reference may be made to the above embodiments, and details are not described in the following embodiments.

(35) FIG. 5 is a cross-sectional schematic view of a light emitting device **10A** according to an embodiment of the disclosure.

(36) The light emitting device **10A** of FIG. 5 is similar to the light emitting device **10** of FIG. 1, with the difference between the two is that the light emitting device **10A** of FIG. 5 further includes an optical microstructure **180** disposed on the transmission path of a light beam L emitted by the

active layer **130**.

(37) It is worth mentioning that the light emission range of the active layer **130** may be controlled by adjusting the reverse bias voltage of the third electrode **160**. Using the third electrode **160** to control the light emission range of the active layer **130** and the secondary optical effect of the optical microstructure **180**, the light emitting device **10A** may have a special light field. The following is an example for illustration with FIG. 5.

(38) Referring to FIG. 5, in this embodiment, the active layer **130** has a first region **R1** and a second region **R2**, and the second region **R2** of the active layer **130** is located between the first region **R1** of the active layer **130** and the side wall **Sc** of the semiconductor structure **S**. The first region **R1** is the middle region of the active layer **130**. The second region **R2** is the peripheral region of the active layer **130**, and the optical microstructure **180** of the light emitting device **10A** includes a first optical microstructure **181** overlapping the active layer **130**. For example, in this embodiment, the third electrode **160** is applied with a reverse bias voltage, and the electric field caused by the third electrode **160** may concentrate the light emission range of the active layer **130** in the first region **R1** (i.e., the middle region), thus the luminous efficiency of the light emitting device **10A** is high. The first region **R1** of the active layer **130** emits a light beam **L**, and the first optical microstructure **181** may converge the light beam **L**, thereby the light emitting device **10A** may provide a special light field concentrated in the front view direction. In this embodiment, the first optical microstructure **181** may be a convex lens that protrudes toward a direction **z** away from the active layer **130**, but the disclosure is not limited thereto.

(39) FIG. 6 is a cross-sectional schematic view of a light emitting device **10B** according to an embodiment of the disclosure. The light emitting device **10B** of FIG. 6 is similar to the light emitting device **10A** of FIG. 5, and the difference between the two is that the optical microstructure **180** of FIG. 6 is different from the optical microstructure **180** of FIG. 5.

(40) Referring to FIG. 6, specifically, in this embodiment, the optical microstructure **180** of the light emitting device **10B** includes a second optical microstructure **182**, which overlaps the active layer **130**. For example, in this embodiment, the third electrode **160** is applied with a reverse bias voltage, and the electric field caused by the third electrode **160** may concentrate the light emission range of the active layer **130** in the first region **R1** (i.e., the middle region), thus the luminous efficiency of the light emitting device **10B** is high. The first region **R1** of the active layer **130** emits a light beam **L**, and the second optical microstructure **182** overlapping the active layer **130** may diverge the light beam **L**, thereby the light emitting device **10A** may provide a special light field concentrated in the side view direction. In this embodiment, the second optical microstructure **182** may be a prism structure, but the disclosure is not limited thereto.

(41) FIG. 7 is a cross-sectional schematic view of a light emitting device **10C** according to an embodiment of the disclosure. In particular, FIG. 7 shows a situation in which the light beam **L** is transmitted without biasing the third electrode **160**.

(42) FIG. 8 is a cross-sectional schematic view of a light emitting device **10C** according to an embodiment of the disclosure. In particular, FIG. 8 shows a situation in which the light beam **L** is transmitted when the third electrode **160** has a reverse bias.

(43) The light emitting device **10C** of FIG. 7 and FIG. 8 is similar to the light emitting device **10A** of FIG. 5, and the difference between the two is that the optical microstructure **180** of FIG. 7 and FIG. 8 is different from the optical microstructure **180** of FIG. 5. Referring to FIG. 7 and FIG. 8, specifically, in this embodiment, the optical microstructure **180** includes a first optical microstructure **181** and a second optical microstructure **182**, and the first optical microstructure **181** and the second optical microstructure **182** overlap the first region **R1** and the second region **R2** of the active layer **130**, respectively. The first optical microstructure **181** converges the light beam **L**, and the second optical microstructure **182** diverges the light beam **L**.

(44) In the embodiment of FIG. 7, the third electrode **160** of the light emitting device **10C** may not be biased, and the first region **R1** (i.e., the middle region) and the second region **R2** (i.e., the

peripheral region) of the active layer **130** may both emit a light beam **L**. The light beam **L** from the first region **R1** may pass through the first optical microstructure **181** in the front view direction, and the light beam **L** from the second region **R2** may pass through the second optical microstructure **182** in the side view direction. Thereby, the light emitting device **10C** may provide a uniform light field in all directions.

(45) In the embodiment shown in FIG. **8**, the third electrode **160** of the light emitting device **10C** may have a reverse bias, and the electric field caused by the third electrode **160** may concentrate the light emission range of the active layer **130** in the first region **R1** (i.e., the middle region). The light beam **L** emitted from the first region **R1** of the active layer **130** may pass through the first optical microstructure **181** and be transmitted in the front view direction. Thereby, the light emitting device **10C** may provide a light field concentrated in the front view direction.

(46) Referring to FIG. **7** and FIG. **8**, by adjusting the bias voltage of the third electrode **160**, the first optical microstructure **181**, and the second optical microstructure **182**, the light field of the same light emitting device **10C** may be changed, thereby the display apparatus (not shown) including the light emitting device **10C** may be switched between different modes and applied in a specific field.

(47) For example, in this embodiment, when the bias voltage of the third electrode **160** of the light emitting device **10C** is substantially 0, the light emitting device **10C** provides a uniform light field in all directions, thereby the display apparatus including the light emitting device **10C** may operate in a sharing mode; when the third electrode **160** of the light emitting device **10C** has a reverse bias, the light emitting device **10C** provides a light field concentrated in the front view direction, thereby the display apparatus including the light emitting device **10C** may operate in a privacy mode. However, the disclosure is not limited thereto, and in other embodiments, the display apparatus including the light emitting device **10C** may also be applied in other fields, such as but not limited to: a three-dimensional display field.

(48) FIG. **9** is a cross-sectional schematic view of a light emitting device **10D** according to an embodiment of the disclosure.

(49) The light emitting device **10D** of FIG. **9** is similar to the light emitting device **10** of FIG. **1**, the difference between the two is that the positions of the first conductive pattern **171**, the second conductive pattern **172**, and the third conductive pattern **173** of the light emitting device **10D** of FIG. **9** are different from the positions of the first conductive pattern **171**, the second conductive pattern **172**, and the third conductive pattern **173** of the light emitting device **10** of FIG. **1**.

(50) Referring to FIG. **9**, specifically, in this embodiment, after the light emitting device **10D** is transposed and electrically connected to the driving backplane (not shown), the semiconductor structure **S** of the light emitting device **10D** is disposed between the first conductive pattern **171** and the driving backplane, between the second conductive pattern **172** and the driving backplane, and between the third conductive pattern **173** and the driving backplane. In other words, in this embodiment, the light emitting device **10D** may be a lateral light emitting diode (lateral LED).

(51) FIG. **10** is a cross-sectional schematic view of a light emitting device **10E** according to an embodiment of the disclosure. The light emitting device **10E** of FIG. **10** is similar to the light emitting device **10D** of FIG. **9**, the difference between the two is that the light emitting device **10E** of FIG. **10** further includes a first optical microstructure **181** overlapping the active layer **130**.

(52) FIG. **11** is a cross-sectional schematic view of a light emitting device **10F** according to an embodiment of the disclosure. The light emitting device **10F** of FIG. **11** is similar to the light emitting device **10D** of FIG. **9**, the difference between the two is that the light emitting device **10F** of FIG. **11** further includes a second optical microstructure **182** overlapping the active layer **130**.

(53) FIG. **12** is a cross-sectional schematic view of a light emitting device **10G** according to an embodiment of the disclosure. The light emitting device **10G** of FIG. **12** is similar to the light emitting device **10D** of FIG. **9**, the difference between the two is that the light emitting device **10G** of FIG. **12** further includes a first optical microstructure **181** and a second optical microstructure

182, which overlap the first region **R1** and the second region **R2** of the active layer **130**, respectively.

(54) FIG. **13** is a cross-sectional schematic view of a light emitting device **10H** according to an embodiment of the disclosure. The light emitting device **10H** of FIG. **13** is similar to the light emitting device **10** of FIG. **1**, the difference between the two is that: in the embodiment of FIG. **13**, the first electrode **151** and the second electrode **152** of the light emitting device **10H** are respectively disposed on different sides of the active layer **130**, and the light emitting device **10H** is a vertical light emitting diode (vertical LED). In addition, the light emitting device **10H** further includes an insulating layer **190**, the insulating layer **190** is disposed on the third electrode **160**, and the third conductive pattern **173** may fill an opening **190a** of the insulating layer **190** to be electrically connected to the third electrode **160**.

(55) FIG. **14** is a cross-sectional schematic view of a light emitting device **10I** according to an embodiment of the disclosure. The light emitting device **10I** of FIG. **14** is similar to the light emitting device **10H** of FIG. **13**, the difference between the two is that: in the embodiment of FIG. **13**, the third conductive pattern **173** and the second conductive pattern **172** are disposed on the same surface of the semiconductor structure **S** (i.e., on the first surface **Sa**); however, in the embodiment of FIG. **14**, the third conductive pattern **173** and the first conductive pattern **171** are disposed on the same surface of the semiconductor structure **S** (i.e., on the first surface **Sa**).

Claims

1. A light emitting device, comprising: a semiconductor structure, comprising: a first type semiconductor layer, the first type semiconductor layer comprising a first sub-side wall and a second sub-side wall, wherein the first and second sub-side walls are located on different planes; a second type semiconductor layer; and an active layer, disposed between the first type semiconductor layer and the second type semiconductor layer; an insulating layer, disposed on the semiconductor structure; a first electrode, electrically connected to the first type semiconductor layer, wherein the first electrode is disposed on the first and second sub-side walls of the first type semiconductor layer; a second electrode, electrically connected to the second type semiconductor layer; and a third electrode, electrically independent from the first electrode and the second electrode, wherein the insulating layer is located between the third electrode and the semiconductor structure, the first electrode, the second electrode, and the third electrode are structurally separated, the third electrode at least has a first portion and a second portion that are electrically connected to each other, and the first portion of the third electrode is disposed on a side wall of the semiconductor structure, while the second portion of the third electrode is disposed between the first electrode and the second electrode.

2. The light emitting device according to claim 1, wherein the side wall of the semiconductor structure comprises a side wall of the active layer, and the first portion of the third electrode is disposed on the side wall of the active layer.

3. The light emitting device according to claim 2, wherein the side wall of the semiconductor structure further comprises a side wall of the first type semiconductor layer, and the first portion of the third electrode is further disposed on the side wall of the first type semiconductor layer.

4. The light emitting device according to claim 3, wherein the side wall of the semiconductor structure further comprises a side wall of the second type semiconductor layer, and the first portion of the third electrode is further disposed on the side wall of the second type semiconductor layer.

5. The light emitting device according to claim 1, further comprising: a first conductive pattern, disposed on the first electrode and is electrically connected to the first electrode; a second conductive pattern, disposed on the second electrode and is electrically connected to the second electrode; and a third conductive pattern, disposed on the third electrode and is electrically connected to the first portion of the third electrode, wherein the first conductive pattern, the second

conductive pattern, and the third conductive pattern are structurally separated.

6. The light emitting device according to claim 5, wherein the semiconductor structure has a first surface and a second surface opposite to each other, the side wall of the semiconductor structure is disposed between the first surface and the second surface; wherein the second portion of the third electrode is disposed on the first surface of the semiconductor structure; the third conductive pattern is disposed on the second portion of the third electrode.

7. The light emitting device according to claim 1, further comprising: at least one first optical microstructure, overlapping the active layer.

8. The light emitting device according to claim 7, wherein the active layer emits a light beam, and the at least one first optical microstructure converges the light beam.

9. The light emitting device according to claim 7, wherein the active layer has a first region and a second region, the second region of the active layer is located between the first region of the active layer and the side wall of the semiconductor structure, and the light emitting device further comprises: at least one second optical microstructure, wherein the at least one first optical microstructure and the at least one second optical microstructure respectively overlap the first region and the second region of the active layer.

10. The light emitting device according to claim 9, wherein the active layer emits a light beam, the at least one first optical microstructure converges the light beam, and the at least one second optical microstructure diverges the light beam.

11. The light emitting device according to claim 1, further comprising: at least one second optical microstructure, overlapping the active layer, wherein the active layer emits a light beam, and the at least one second optical microstructure diverges the light beam.

12. The light emitting device according to claim 1, wherein the third electrode has a reverse bias.
